

Pattern Discovery in U.S. Transportation Systems

CS 4412: Data Mining – M4 Final Deliverable
Using Monthly Transportation Statistics

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Abstract

This study investigates latent structure in the U.S. Monthly Transportation Statistics dataset to characterize relationships between transportation activity and safety outcomes across multiple modes.

An integrated analytical framework is employed, combining unsupervised clustering (K-Means), dimensionality reduction (Principal Component Analysis), interpretable classification (Decision Trees), and temporal analysis. Cluster selection is guided by silhouette analysis, which favors $k = 7$ (0.51), while $k = 3$ is selected to balance statistical separation with interpretability.

The results identify three distinct transportation system profiles differentiated by activity intensity and safety performance. Notably, one cluster exhibits high activity alongside comparatively lower fatality rates, challenging the assumption that increased utilization necessarily leads to elevated risk.

Temporal analysis further reveals that cluster membership is strongly aligned with time, indicating that observed patterns reflect not only structural differences but also the historical evolution of transportation systems.

This study demonstrates that combining clustering, interpretability, and temporal context provides a robust framework for analyzing complex transportation systems.

1 Introduction

Understanding the relationship between transportation activity and safety outcomes is a central challenge in transportation analysis. These systems operate across multiple modes and involve complex interactions between infrastructure, demand, and risk.

This study addresses the following discovery questions:

- Do distinct transportation system profiles emerge?
- How does activity relate to safety outcomes?
- Which variables drive system differences?
- Are patterns structural or temporal?

The dataset was transitioned from an automobile-focused dataset (M1) to the Monthly Transportation Statistics dataset (M2) to support richer, multi-modal analysis.

2 Methodology

2.1 Preprocessing

- Removed sparse features (<50% coverage)
- Replaced infinite values with NaN
- Applied mean imputation
- Standardized features using z-score normalization

2.2 Cluster Selection

Silhouette analysis was used to evaluate clustering performance across different values of k . The highest score was achieved at $k = 7$ (approximately 0.51), indicating stronger numerical separation. However, $k = 3$ was selected for the final model due to its interpretability and ability to produce clearly distinguishable system profiles.

This tradeoff reflects a deliberate choice to prioritize meaningful interpretation over marginal improvements in clustering metrics.

2.3 Modeling Approach

- K-Means clustering ($k = 3$)
- PCA visualization
- Decision Tree interpretation
- DBSCAN (alternative clustering)

3 Results

3.1 Cluster Interpretation

The clustering results reveal three distinct transportation system profiles that differ in both activity levels and safety outcomes.

Cluster 0: Elevated Risk, Moderate Activity

Higher rail and aviation fatalities despite moderate activity suggest inefficiencies or infrastructure limitations.

Cluster 1: High Activity, Lower Risk

This cluster exhibits the highest activity with lower fatality rates. This contradicts expected patterns and suggests improved infrastructure or safety systems.

Cluster 2: Transitional Systems

Moderate activity and moderate risk indicate systems undergoing change or partial improvement.

3.2 PCA Visualization

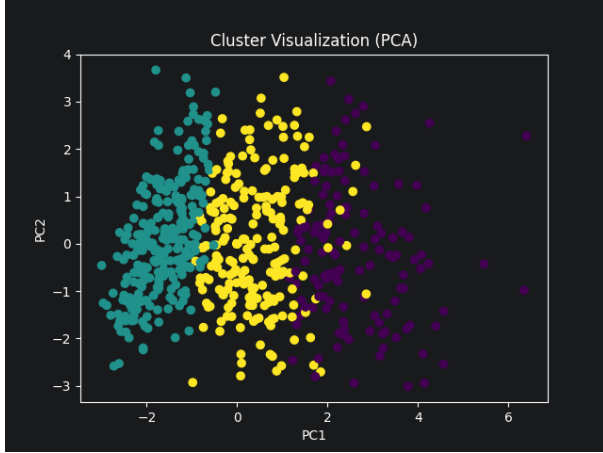


Figure 1: PCA projection of clustered data

Clusters are clearly separable, supporting clustering validity.

3.3 Temporal Analysis

Analysis of cluster membership over time indicates that clusters are not randomly distributed across the dataset. Earlier observations tend to align with higher-risk profiles, while more recent observations correspond to lower-risk, higher-efficiency systems.

This suggests that the identified clusters reflect not only structural differences, but also the historical evolution of transportation systems, including improvements in infrastructure, regulation, and safety practices.

3.4 Decision Tree Interpretation

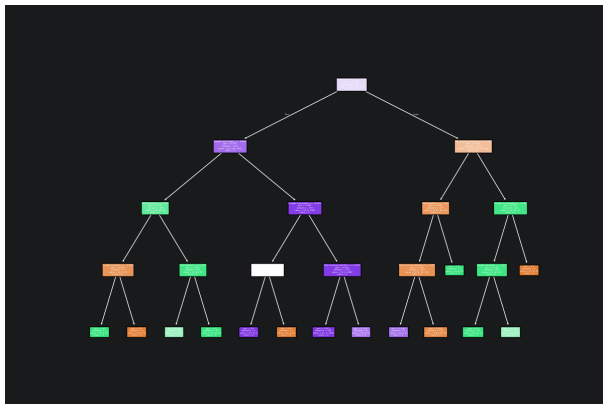


Figure 2: Decision tree for cluster classification

Rail fatalities (approximately a threshold of 76) form the primary split, with aviation fatalities further refining

classification. These results indicate that safety-related variables are the dominant drivers of cluster separation.

3.5 Alternative Clustering

DBSCAN did not produce meaningful clusters, suggesting the data is better suited for partition-based methods such as K-Means.

4 Validation

K-Means was run across multiple random seeds. Silhouette scores remained consistent, indicating that the clustering structure is stable and not sensitive to initialization.

5 Interpretation

Transportation systems differ not only in activity levels but in their ability to manage risk.

A key finding is that high activity does not necessarily increase fatalities. Instead, modern systems achieve higher throughput while maintaining improved safety outcomes.

Temporal analysis suggests that these improvements reflect long-term system evolution driven by infrastructure, policy, and technological advancements.

6 Critical Assessment

6.1 Validity

Results are supported by silhouette scores, PCA separation, and stability testing, though the absence of ground truth labels limits definitive validation.

6.2 Limitations

- Missing data required imputation
- K-Means assumptions (spherical clusters)
- Sensitivity to feature selection

6.3 Temporal Confounding

Cluster structure is partially influenced by time, limiting causal interpretation.

6.4 Ethical Considerations

Care must be taken to avoid overgeneralizing safety conclusions or inferring causality from correlational patterns.

7 Conclusion

This study identifies three transportation system profiles shaped by both activity levels and safety outcomes.

The results demonstrate that increased activity does not inherently increase risk, highlighting the importance of infrastructure, regulation, and system efficiency.

Temporal analysis indicates that these patterns reflect long-term evolution in transportation systems.

Future work includes exploring alternative clustering methods and more advanced temporal modeling techniques.